Mystery Basket Scope

Building AI-Powered Customer
Insights with Transparency & Trust

Woolworths × Quantium

Scope & Objective

Build a scalable system to simulate real customer shopping and compare basket prices across competitors.

Scope (MVP)

- 8 shopping missions × 5 personas × 3 retailers = 120 automated shops
- Focus: price, promotions, and realistic customer choice behaviour
- Output: Comparable baskets + actionable pricing insights

What this enables

- Consistent, repeatable competitive intelligence
- Customer-segment view of value (not just average pricing)
- Foundation for ongoing monitoring and optimisation

Key constraint (transparent upfront)

- Frequency depends on data refresh quality (not real-time scraping)

Approach & Why It Works

What we learned from the POC

- Pure LLM approach isn't best for production: unreliable outputs, Unreliable SKU Matching, no product grounding, weekly barries from retailers for bot protection

Our approach (robust by design)

1. Structured Attribute Layer (critical foundation)
    - Standardised product data across retailers
    - Includes price, promotions, quality markers, brand, origin, etc.
    - Ensures decisions are based on real, comparable inputs
2. Persona-Based Decision Engine
    - Each persona uses different signals:
        - Saver → promotions & price
        - Conscious → quality & ethics
        - Traditional → trusted brands
    - Produces realistic, differentiated baskets
3. Validation & Comparison Layer
    - Ensures basket completeness and pricing accuracy
    - Outputs clear WW vs Coles vs Aldi comparisons

Why this works
Moves from "AI guessing" to data-grounded simulation of real behaviour

How We Enable Persona-Based Decisions

Raw product data isn't enough, competitor datasets lack the attributes needed to simulate real customer choices (e.g. organic, free-range, trusted brands)

Our Approach: Build a Standardised Attribute Matrix
We create a unified product layer across all retailers, where every product is enriched with consistent decision signals:

- Price & Promotions → price, unit price, discount type
- Quality & Ethics → organic, free-range, sustainability markers
- Brand Signals → own brand, trusted brand, premium tier
- Product Details → pack size, ingredients, nutrition

→ This ensures all products are comparable and decision-ready

How the System Uses It

- Retrieve (RAG layer)
- Pull relevant products per ingredient with full attribute set
- Decide (AI agents)
- Multiple agents select products based on persona rules

e.g. Saver → price & promotions | Conscious → quality & ethics

- Validate (Consistency layer)
- Cross-check basket completeness, balance, and pricing accuracy

Why This Matters

Enables realistic, differentiated customer behaviour

Moves from price-only comparisons → true customer simulation

Creates a scalable, reusable foundation for all future use cases

Delivery timeline

Week 1-Phase 1: Foundation & Data
27 Apr

⚠️ Public holiday Monday and innovation day Friday ⚠️

- Data assessment ✅
- Product catalog build (Woolworths, Coles, Aldi)✅
- Aldi enrichment (~2000 products)✅

Week 2 -Phase 2: LLM Decision Engine
4 May

- Set up ADK
- Build persona-based agent template
- Implement -agent voting mechanism per ingredient
- Implement persona prompts for each CREST segment
- Semantic search (vector embeddings)

Week 3 -Phase 3: Scale Up
11 May

- Build LLM judge for basket validation
- Implement re-run logic for failures
- Build Pipeline
- Test full pipeline: RAG → Voting → Judge

Week 4 -Phase 4: Production Ready
18 May

- Run all 8 missions × 5 personas × 3 retailers
- Build front-end interface (basket, price, justification, rankings)
- Analytics and comparison tables
- Documentation and handoff

Week 5
25 May

Decisions required for sign-off

Pricing approach

- Use single representative postcode for competitor pricing (average across all stores for competitors?)
  → Ensures consistency and real-world comparability
- For Woolies, do we go for average price across nsw for competitors or limit to 1 store

Averaged across NSW metropolitan

Execution model

- Run on data refresh cadence (e.g. weekly) vs daily scraping
  → Prioritises accuracy and stability

Weekly

Do we want production level deployment? (this will lead to extended project time)

- Custom domain name
- dev>test>uat>prod pipeline build and testing

Non-prod for faster delivery

CREST Segment Attributes needed

Main Personas with Different Needs:

1. CONSCIOUS (17%) - Young, health & sustainability focused
- Critical attributes: organic, free_range, rspca_approved, grass_fed, msc_certified, gluten_free, vegan, sustainable
- Willing to pay MORE for values alignment

2. REFINED (15%) - Affluent, older, quality over price
- Critical attributes: premium, gourmet, well_known_brand, specialty, fresh_unprocessed
- Quality is 61% priority vs only 21% for price

3. ESSENTIAL (17%) - Singles/couples, price conscious
- Critical attributes: price, own_brand, value_range, small_pack, convenience
- Prioritize price over quality, prepared to shop around

4. SAVERS (23% - LARGEST) - Young families, tight budget
- Critical attributes: price, unit_price, promotion_type (Yellow/Red), pack_size, family_size
- Highest engagement with promotions and value hacks
- Unit price CRITICAL for comparison

5**. TRADITIONAL (28% - 2ND LARGEST)** - Mature, seek familiarity
- Critical attributes: familiar_brand, australian_made, tried_tested, quality, fresh_ingredients
- Stick to brands they know and trust